
Police dog training
COMPLEX
100 Years
Wall of Conquerors
40 40 40 40
40 40 40
don't Quit
ICD
Beware
VTACH DOG
BRUTAL!
EGAD!
LYTES
High
Voltage
CPR
ANTI-DRONE TRAINING
PRO CANINE
AMIGO
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44

2.4 - Wide-Complex Tachycardia SOAP

1(O) ECG grid fence:
Tachycardia (≥ 100 bpm) on ECG → wide complex? (QRS ≥ 120ms) → is rhythm regular (monomorphic VT, SVT with aberrancy, SVT with pre-excitation, antidromic AVRT) or irregular? (AF with aberrancy, AF with pre-excitation, polymorphic VT)

2(O) “100” years:
**Tachycardia (≥ 100 bpm)** on ECG → wide complex? (QRS ≥ 120ms) → is rhythm regular (monomorphic VT, SVT with aberrancy, SVT with pre-excitation, antidromic AVRT) or irregular? (AF with aberrancy, AF with pre-excitation, polymorphic VT)

3(O) Jump over three “40” boxes:
Tachycardia (≥ 100 bpm) on ECG → **wide complex? (QRS ≥ 120ms)** → is rhythm regular (monomorphic VT, SVT with aberrancy, SVT with pre-excitation, antidromic AVRT) or irregular? (AF with aberrancy, AF with pre-excitation, polymorphic VT)

4) Wide “C O M P L E X”:
wide complex tachycardia (QRS ≥ 130ms) → **ventricles activated outside normal conduction system** (e.g. SVT with pre-excitation, antidromic SVT); **conduction system abnormality** (SVT with aberrancy); **impulse originates in ventricle** (v tach)

5(P) ACLS cart toward unstable patient with broken heart:
patient with wide complex tachycardia in **cardiac arrest** (or who becomes pulseless) → **ACLS protocol** (e.g. CPR and immediate high-energy countershock)

6(O) Broken heart:
all wide complex tachycardias can lead to **hemodynamic compromise, cardiac arrest, and sudden cardiac death** → most likely in setting of v tach/ v fib (especially in patients with underlying structural heart disease, e.g. CAD with previous MI)

7) Broken heart structure underneath:
all wide complex tachycardias can lead to hemodynamic compromise, cardiac arrest, and sudden cardiac death → most likely in setting of v tach/ v fib (**especially in patients with underlying structural heart disease, e.g. CAD with previous MI**)

8(P) Synchronized heart watches + paddles next to unstable but responsive guy:
**unstable** patients with wide complex tachycardia who are **responsive and +pulse** → immediate **synchronized cardioversion**

9(O) Fainting guy with chest pain + ↓brain helmet + failing heart balloon:
patient with wide complex tachycardia → assess for hemodynamic instability (**AMS, c/p, hypotension, syncope, or signs of HF**)

10(O) ==P== Palpating chest:
**stable** patients with wide complex tachycardia typically present with **palpitations**/ sense of tachycardia → put on continuous cardiac monitoring (can deteriorate rapidly) → inspect ECG/history for etiology

11(O) Regular vs irregular notes:
Tachycardia (≥ 100 bpm) on ECG → wide complex? (QRS ≥ 120ms) → is rhythm **regular**(monomorphic VT, SVT with aberrancy, SVT with pre-excitation, antidromic AVRT) or **irregular?** (AF with aberrancy, AF with pre-excitation, polymorphic VT)

12) "#1" above "vtach" dog:
**ventricular tachycardia is the #1 cause of wide complex tachycardia** (WCT) → suspect v tach in patients with WCT of unknown etiology who are >35yo +/- hx of heart dz (esp CAD w/ hx of MI or angina, CHF) → check prior ECG for signs of infarct

13(O) Deadly wide monomorphic teeth:
**sustained (>30sec) monomorphic v tach (QRSs may be slightly variable but wide/stable/regular rhythm)** → associated with underlying structural heart disease (e.g. CAD with previous MI, cardiomyopathy) → #1 cause (along with v fib) of sudden cardiac arrest/ sudden cardiac death

14) Coronary crown next to "vtach" dog:
sustained monomorphic v tach → **associated with underlying structural heart disease**(e.g. **CAD with previous MI**, cardiomyopathy) → #1 cause (along with v fib) of sudden cardiac arrest/ sudden cardiac death

15) Dilated dog treat bag next to "vtach" dog:
sustained monomorphic v tach → **associated with underlying structural heart disease**(e.g. CAD with previous MI, **cardiomyopathy**) → #1 cause (along with v fib) of sudden cardiac arrest/ sudden cardiac death

16) "#1" finger pointing from "vtach" dog to guy in SCA/SCD:
sustained monomorphic v tach → associated with underlying structural heart disease (e.g. CAD with previous MI, cardiomyopathy) → **#1 cause (along with v fib) of sudden cardiac arrest/ sudden cardiac death**

17(O) “Vtach” dog trainer with “30” timer + monitoring binoculars:
**nonsustained monomorphic v tach (<30sec)** → typically **asymptomatic** and picked up on continuous monitor → **generally benign if pt has no underlying structural heart disease → can lead to SCD in pts w/ underlying cardiac ischemia/ HF!** → assess for condition causing irritability of ventricular myocardium (e.g. hypoxia, hypoK+, hypoMg, myocardial ischemia, or decompensated CHF, etc.)

18(O) Atrial dog signals above bear keychain:
**SVT with aberrancy/ pre-excitation** → SVT (e.g. sinus tach, AT, A flutter, AVNRT, etc.) encounters an abnormality in conduction system (e.g. bundle branch block) or accessory pathway (ventricular “pre-excitation” with every beat) → **regular wide complex tachycardia (irregular if A fib)**

19) SVT dog encountering bundle:
SVT with aberrancy = **SVT (e.g. sinus tach, AT, A flutter, AVNRT, etc.) encounters abnormality in conduction system, e.g. bundle branch block** (or “functional block” appears only when patient is in tachycardia → rapid impulses don’t let fibers recover) → **regular wide complex tachycardia** (irregular if A fib)
Symbols

20) SVT dog exciting wolf on extra path:
SVT with pre-excitation = **SVT (e.g. sinus tach, AT, A flutter, AVNRT, etc.) encounters encounters accessory pathway** (e.g. in patient with **WPW** → ventricular pre-excitaiton with every beat) → **regular wide complex tachycardia** (irregular if A fib)

21) “Anti-drone” SVT dog jumping down extra path:
**antidromic AVRT (rare)** → reentrant circuit travels up through AV node to atria → down accessory pathway to ventricles → **regular wide complex tachycardia**

22(O) Irregular wide complex fence:
**Polymorphic ventricular tachycardia = >100 bpm + continuously varying QRS morphology (irregular rate)** → related to acquired/congenital long QT (“torsades de pointes”), or structural heart disease (e.g. CAD), or familial → can cause hemodynamic instability, SCA/SCD (but commonly spontaneously terminates)

23) “don’t QuiT” with twisted straps:
**“torsades de pointes”** (polymorphic ventricular tachycardia) → related to acquired (e.g. antiarrhythmic drugs, some antimicrobials, etc.) or congenital **long QT** → can cause hemodynamic instability, SCA/SCD (but commonly spontaneously terminates)

24(O) Irregularly irregular static screen above bundle and extra path:
**A fib with aberrancy/ pre-excitation** → A fib signals encounter an abnormality in conduction system (e.g. bundle branch block) or accessory pathway (ventricular "pre-excitation" with every beat) → **irregular wide complex tachycardia**

25(O) "Vtach" dog jumping over 4 boxes:
Patient with regular wide complex tachycardia of unknown etiology → V tach is distinguished by **especially wide QRSs (>140 - >160ms)**, extreme axis deviation, concordance, AV dissociation on ECG

26(O) Extremely tilted post:
Patient with regular wide complex tachycardia of unknown etiology → V tach is distinguished by especially wide QRSs, **extreme axis deviation** ("NW" -90 to ±180; OR ±40 form baseline; OR left of -30 in RBBB-like WCT; OR right of +90 in LBBB-like WCT), concordance, AV dissociation on ECG

27(O) Symmetric wall of "conquerors 1-6":
Patient with regular wide complex tachycardia of unknown etiology → V tach is distinguished by especially wide QRSs, extreme axis deviation, **concordance in precordial leads V1-6 (QRS complexes are monophasic and same polarity)**, AV dissociation on ECG

28(O) Dissociated "P" squirrels on QRS spike fence:
Patient with regular wide complex tachycardia of unknown etiology → V tach is distinguished by especially wide QRSs, extreme axis deviation, concordance, **AV dissociation (e.g dissociated P waves, PP interval ≠ RR interval)** on ECG

29(O) Capturing "vtach" dog with QRS collar:
Patient with regular wide complex tachycardia of unknown etiology → V tach is distinguished by especially wide QRSs, extreme axis deviation, concordance, **AV dissociation (e.g fusion beats= ventricular impulse coincides with SA-derived impulse → intermediate width QRS; capture beats = regular conduction pathway → narrow QRS)** on ECG

30(O) Previous ECG files:
Patient with wide complex tachycardia of unknown etiology → **inspect previous/ baseline ECG** → History of SVT? Fascicular and/or bundle branch block causing aberrant conduction? Prior infarction causing v tach? Ventricular pre-excitation/ WPW? Brugada causing v tach? Long QT causing torsades? → patients with A fib (irregular) baseline rhythm but now in regular WCT = v tach

31(S) Old police chief with “vtach” dogs:
ventricular tachycardia is the #1 cause of wide complex tachycardia (WCT) → **suspect v tach in patients with WCT of unknown etiology who are >35yo** +/- hx of heart dz (esp CAD w/ hx of MI or angina, CHF) → younger healthy pts more likely to have SVT (**Demographics**)

32(S) ==O== Clogged red coronary crown:
ventricular tachycardia is the #1 cause of wide complex tachycardia (WCT) → suspect v tach in patients with WCT of unknown etiology who are >35yo **+/- hx of heart dz (esp CAD w/ hx of MI or angina, CHF) → check prior ECG for signs of infarct (PMHx/ECG)**

33(S) “ICD” vest:
ventricular tachycardia is the #1 cause of wide complex tachycardia (WCT) → suspect v tach in patients with WCT of unknown etiology who have an ICD already placed (carry known risk of v tach/ v fib) (**PMHx**)

34(S) ==P==Magnet badge:
Pacemaker present? → consult cardiology for placement of a magnet over the pacemaker to distinguish device-associated WCT (**PMHx/Plan**)

35(O) “Brutal! Egad!”:
Patient with wide complex tachycardia of unknown etiology → inspect previous/ baseline ECG → **Brugada syndrome**? (autosomal dominant cardiac sodium channel disorder = pseudo-RBBB + ↑ST in leads V1 to V2) → ↑risk of v tach, SCA/SCD!

36(S) ==O== History with SVT companion:
Patient with wide complex tachycardia of unknown etiology → **get history/ inspect baseline ECG → History of SVT with wide complex? Fascicular and/or bundle branch block causing aberrant conduction? Ventricular pre-excitation/ WPW (delta wave + ↓PR)?** → patients with A fib (irregular) baseline rhythm but now in regular WCT = v tach (**PMHx/ECG**)

37(S) ==O== “don’t QuiT” with med bag:
“torsades de pointes” (polymorphic ventricular tachycardia) → **related to acquired (e.g. antiarrhythmic drugs, some antimicrobials, etc.) or congenital long QT** → obtain medication history + prior ECG (**PMHx/Meds/ECG**)

38(O) Inspecting electrolyte water:
Patient with wide complex tachycardia → check electrolytes (**hypokalemia and hypomagnesemia can cause V tach**)

39) T-bone:
Patient with wide complex tachycardia → **check cardiac troponins** (especially in patients presenting with chest pain and hemodynamic instability) → MI causing v tach?

40(P) Grabbing SVT dog neck:
STABLE patient with wide complex tachycardia and suspected SVT with aberrant conduction can initially be treated with a **valsalva maneuver or carotid massage** → if this is unsuccessful, adenosine can be both therapeutic and diagnostic (AV node block terminates AVRT/AVNRT, or slow rate enough to expose A tachy P waves/ A flutter waves/ A fibrillation → proceed down respective tx algorithm)

41(P) Dancing dog:
STABLE patient with wide complex tachycardia and suspected SVT with aberrant conduction can initially be treated with a valsalva maneuver or carotid massage → if this is unsuccessful, **adenosine can be both therapeutic and diagnostic** (AV node block terminates AVRT/AVNRT, or slow rate enough to expose A tachy P waves/ A flutter waves/ A fibrillation → proceed down respective tx algorithm); CAUTION adenosine can rarely cause hemodynamic collapse → ensure resuscitation equipment is available

42(P) “Amigo” the “pro canine” with paddles handy:
STABLE patient with wide complex tachycardia and suspected ventricular tachycardia can be treated either **IV amiodarone, or IV procainamide** (if no significant hypotension) → refractory WCT? synchronized cardioversion (some clinicians go straight to synchronized cardioversion in the setting of suspected v tach)

43(P) Police dog “magazine”:
Torsades de pointes is treated with IV magnesium sulfate

44(P) Durable ICD vest:
**implantable cardioverter defibrillator** +/- adjunctive antiarrhythmic therapy → secondary prevention of SCD in patients with prior v tach/ v fib, or primary prevention of SCD in patients at ↑risk of life-threatening v tach/ v fib (idiopathic VT may be treated with ablation or meds)